

The associations of diet with serum insulin-like growth factor I and its main binding proteins in 292 women meat-eaters, vegetarians, and vegans.



The lower rates of some cancers in Asian countries than in Western countries may be partly because of diet, although the mechanisms are unknown. The aim of this cross-sectional study was to determine whether a plant-based (vegan) diet is associated with a lower circulating level of insulin-like growth factor I (IGF-I) compared with a meat-eating or lacto-ovo-vegetarian diet among 292 British women, ages 20-70 years. The mean serum IGF-I concentration was 13% lower in 92 vegan women compared with 99 meat-eaters and 101 vegetarians ($P = 0.0006$). The mean concentrations of both serum IGF-binding protein (IGFBP)-1 and IGFBP-2 were 20-40% higher in vegan women compared with meat-eaters and vegetarians ($P = 0.005$ and $P = 0.0008$ for IGFBP-1 and IGFBP-2, respectively). There were no significant differences in IGFBP-3, C-peptide, or sex hormone-binding globulin concentrations between the diet groups. Intake of protein rich in essential amino acids was positively associated with serum IGF-I (Pearson partial correlation coefficient; $r = 0.27$; $P < 0.0001$) and explained most of the differences in IGF-I concentration between the diet groups. These data suggest that a plant-based diet is associated with lower circulating levels of total IGF-I and higher levels of IGFBP-1 and IGFBP-2.